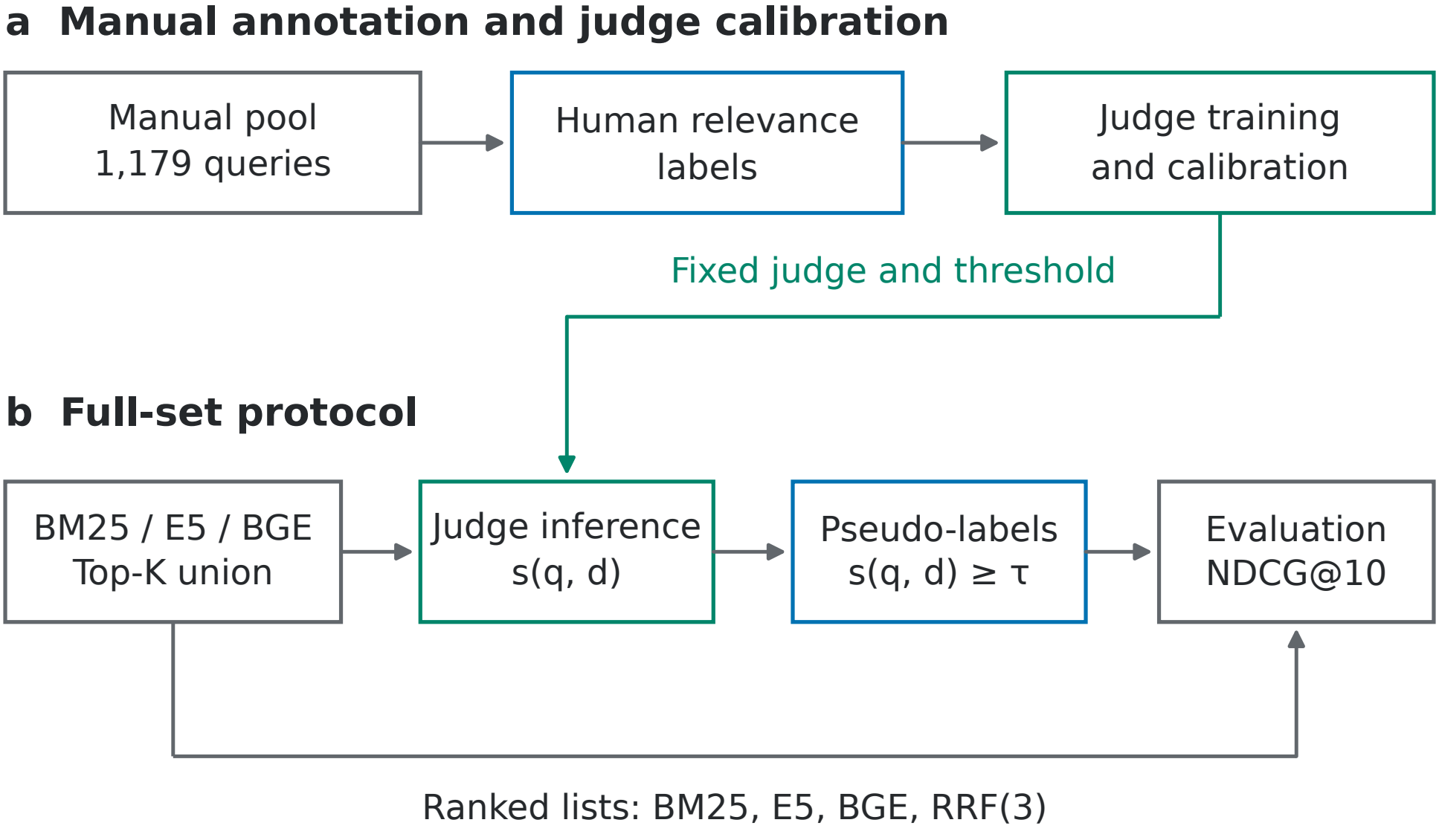
a Manual annotation and judge calibration
Manual pool
1,179 queries
Human relevance
labels
Judge training
and calibration
Fixed judge and threshold
b Full-set protocol
BM25 / E5 / BGE
Top-K union
Judge inference
s(q, d)
Pseudo-labels
s(q, d) ≥ τ
Evaluation
NDCG@10
Ranked lists: BM25, E5, BGE, RRF(3)